ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **WINTER SEMESTER, 2022-2023**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4305: Computer Organization and Architecture

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Categorize various cache types according to their location in relation to the processor, Memory Management Unit (MMU), and main memory. Compare their operational process with the required figures. — 5 (CO1) (PO1)

   b) Consider a 16-bit data 0101000000111001 stored in the main memory. Generate a Hamming SEC-DED code for it that demonstrates each step in between. Additionally, demonstrate that the code accurately detects an error in data bit 5. — 10 (CO1) (PO1)

   c) In pipeline planning, a special technique known as branch prediction can be utilized for predicting whether a branch will be taken to avoid branch hazard. Provide a brief explanation of the various branch predictions techniques along with relevant figures and examples. — 10 (CO1) (PO1)

2. a) Magnetic read-write operations refer to the process by which data is written to and subsequently read from a magnetic disk. Illustrate these mechanisms with the use of suitable figures. Additionally, briefly describe the formatting and organization of data on the platter of a magnetic disk that illustrates the disk's data structure. — 8 + 7 (CO1) (PO1)

   b) Differentiate between the following pairs: — 5 × 2 (CO1) (PO1)
   i. CLV and CAV
   ii. Strip and stripe
   iii. Hamming code and Hamming distance
   iv. Page and frame
   v. PCI and PCIe

3. a) There has been an upward trend of increasing component sophistication and complexity as computer systems have developed. The I/O function is the one place where this is most obvious. Provide an appropriate example to summarize the I/O Function's evolution process. — 5 (CO1) (PO1)

   b) A computer and an external device are connected via an I/O module. Give a brief explanation of an I/O module's main features along with the required figures. — 5 (CO1) (PO1)

   c) Two addressing modes—memory mapped and isolated—are available when the CPU, main memory, and I/O share a common bus. Explain the differences between isolated and memory-mapped I/O using relevant examples. — 5 (CO1) (PO1)

   d) The majority of virtual memory schemes use a unique cache for page table entries, sometimes referred to as a Translation Lookaside Buffer (TLB), to lower the memory access time. Nevertheless, interaction between the virtual memory mechanism and the conventional cache system is required. — 5 × 2 (CO1) (PO1)
   i. Illustrate the operational stages of a TLB using a flowchart.
   ii. Show the interaction between a TLB and a cache using a figure.

4. A part of main memory content of a 16-bit architecture-based computer system is presented in Figure 1 showing different instructions and data in different locations. It has an instruction set having the following instructions with their corresponding opcodes (4-bit):

- *Load AC* from memory (0001)
- Store *AC* to memory (0010)
- *ADD* to *AC* from memory (0101)

300	1940
301	5941
302	2941
...	
...	
940	0003
941	0002

Figure 1: A part of main memory for Question 4.

a) Convert the instructions stored in memory locations 300-302 into instructions that have 1, 2, and 3 operands. With appropriate assumption(s), variables could be used in place of physical addresses. 10 (CO1) (PO1)

b) Considering the diagram depicted in Figure 2, generate the time sequence of micro-operations and the corresponding control signals to demonstrate the operation of the control unit. The control signal terminations are denoted as C_i and represented by a circular symbols in the figure. 15 (CO3) (PO2)

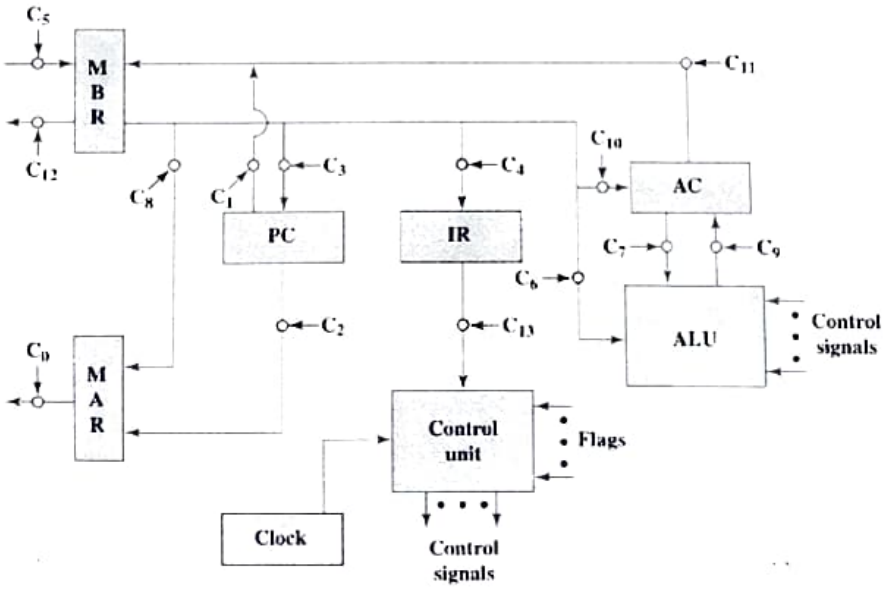


Figure 2: Data paths and control signals for Question 4.b)

c) Demonstrate the events caused in Question 4.b) by utilizing CPU Registers - *IR*, *PC*, *AC*, *MAR*, and *MBR*. Illustrate how their contents are modified over time sequence for all three instructions. 15 (CO3) (PO2)

d) Verify the possibility of different types of hazards in the given scenario when implementing a pipeline strategy. Provide specific examples of these circumstances and propose effective solutions to address these problems. 10 (CO3) (PO2)

5. a) The C structure provided in Code Snippet 1 includes a number of data types. The comments show the content of the corresponding variables. Illustrate the addressing and byte order of big-endian and little-endian layouts on a 64-bit architecture. 10 (CO1) (PO1)

```
1 struct{
2     short i;      //0x1112
3     int j;        //0x13141516
4     double k;     //0x1718192021222324
5     int l;        //
6     char m[7];    //'A','B','C','D','E','F','G'
7 } s;
```

Code Snippet 1: A C structure for Question 5.

b) Consider a byte addressable 16-bit microprocessor where a certain portion of main memory contains data and instruction as depicted in Figure 3. The memory contents from location 401 increase sequentially following the locations 399 – 400. 15 (CO1) (PO1)

200	Load AC	Mode
201	500	
202	Next Instruction	
...		
...		
399	999	
400	1000	

Figure 3: A part of main memory for Question 5.b)

If source register, $R1 = 400$, base register = 100, and index register = 100, determine the Effective Address (EA) and the content of *AC* after executing instruction at address 200 for the following addressing modes:

i. Direct
ii. Indirect
iii. Register
iv. Register indirect
v. Immediate
vi. PC relative
vii. Displacement
viii. Indexing